20 An electric current of 12 A flows in a wire of length 0.35 m and diameter 1.5 mm.

The average drift speed of the free electrons in the wire is 5.0×10^{-4} m s^{-1}.

How many free electrons are there in the wire?

A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}

Ans: B

Since $I = nAve$ and $n = N$ / Volume = N / (A x Length), where N is number of electrons

$I = (N / AL) \times Ave = N \times (ve / L)$

$$N = \frac{IL}{ve} = \frac{(12)(0.35)}{(5.0\times10^{-4})(1.6\times10^{-19})}$$

$$= 5.25\times10^{22} = 5.3\times10^{22}$$